

AE-353: Preparation of samples for ultrafast spectroscopy from [Ru(PNP)] and [Ru(PNN)] in H₂O and D₂O with a rotating stage

Date: 2024-10-07

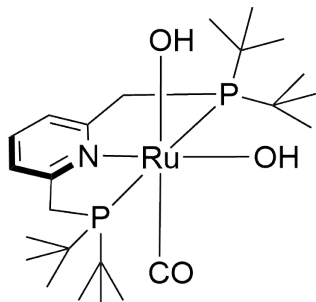
Tags: PNP P(tBu)N(py)P(tBu)
[Ru(CO)(OH)₂(PNP)] AE Ultra Fast
spectroscopy Cooperation AG Dietzek-
Ivanšić

Category: Two-photon

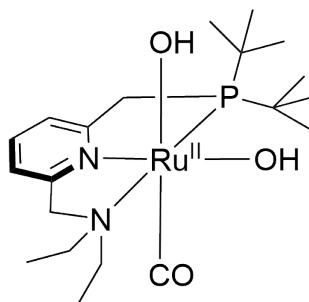
Status: Done

Created by: Alexander Eith

Reaction scheme/sample structure



Chemical Formula: C₂₄H₄₅NO₃P₂Ru
Molecular Weight: 558,64552
[Ru(CO)(OH)₂(PNP)]



Chemical Formula: C₂₀H₃₇N₂O₃PRu
Molecular Weight: 485,57076
[Ru(CO)(OH)₂(PNN)]

Literature/reference experiments

Literature	/
Reproduction	/
Related experiment	AE-325: Preparation of samples for ultrafast and raman spectroscopy from [Ru(CO)(OH)₂(PNPtBu)] (AE-310-2)

Reagents

Name	CAS Number / Experiment Number	Amount [μmol]	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Volume _{theo} [ml]	Volume _{exp} [mL]
[Ru(CO)(OH) ₂ (PNPtBu)]	AE-310: Synthesis of [Ru(CO)(OH)₂(PNPtBu)] using Ag₂O, 5 h (AE-310-2)	2.31 + 2.31	/	1.29 + 1.29	1.24 + 1.27	558.645	/	/
[Ru(CO)(OH) ₂ (PNN)]	AE-343: Synthesis of [Ru(CO)(OH)₂(PNN)] using Ag₂O, 5 h	3.01 + 3.01	/	1.46 + 1.46	1.30 + 1.43	485.571	/	/
Milli-Q water	KRA-104: Degassing of Milli-Q water	/	/	/	/	/	1.0 + 1.0	0.98 + 0.98
D ₂ O	AE-352: Degassing of D₂O	/	/	/	/	/	1.0 + 1.0	0.96 + 0.89

Procedure/observations

All steps (unless stated otherwise) were carried out under argon using standard [Protocol - Schlenk](#)

Technique without heating the glassware prior to use.

Since the compound is light sensitive the steps were carried out with the light turned out in the fumehood and the lab.

Date	Time	Step	Observations
22.09		The obtained cuvettes were cleaned using nitric acid and water	
23.09		Four 10 mL flasks were set under argon	
	9:30	To each flask Ru(PNP) or Ru(PNN) was added. To AE-353-1 1.24 mg Ru(PNP), to AE-353-2 1.27 mg Ru(PNP), to AE-353-3 1.30 mg Ru(PNN) and to AE-353-4 1.43 mg Ru(PNN)	
		The flasks were set under argon	
	10:10	To AE-353-1 D2O (0.96 mL) and to AE-353-3 D2O (0.89 mL) was added.	For -1 a 2.3 mM and for -3 a 3.0 mM mixture is obtained left -1 in D2O, right -3 in D2O.jpg
	10:20	To AE-353-2 H2O (0.98 mL) and to AE-353-4 H2O (0.98 mL) was added.	For -2 a 2.3 mM and for -4 a 3.0 mM mixture is obtained looks analog to -1 and -3
		Each flask was stirred for approx. 10 min	
	10:50	The samples, as well as the cuvettes and all needed equipment was brought to the water glovebox from AG Penvenva (TO 227)	Remember to don't forget anything (think of edding, papers, gloves, the cuvettes etc.)
	11:30	Each sample was transferred into a separate vial using a glass pipette	
	11:40	Each sample was filtered through a PA syringe filter (pore-diameter = 0.20 µm) into a new vial	after filter from left to right -1 -2 -3 -4.jpg
	11:50	Each sample was transferred into a 1 mm cuvette with a PTFE-screw cap	
	12:00	Each cuvette was closed	
	13:00	Under air a 1 mm cuvette with PTFE stopper was filled with D2O (AE-353-5) and another cuvette was filled with H2O (AE-353-6)	

	13:30	The samples were hand over to Mohini from AG Dietzek-Ivansic	
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Results

[Results-and-discussions_RuPNP_202411012.docx](#)

Next steps

Measuring PNN and PNP in H₂O and KOH/H₂O

Linked experiments

- [KRA-104: Degassing of Milli-Q water](#)

- [AE-238: Synthesis of \[Ru\(CO\)\(OH\)₂\(PNPtBu\)\] using Ag₂O, 5 h](#)

- [AE-243: 1 M Stocksolution of KOH in water](#)

- [AE-252: Synthesis of \[Ru\(CO\)\(OH\)₂\(PNPtBu\)\] using Ag₂O, 5 h](#)

- [GD-003: Degassing of Milli-Q-Water](#)

- [AE-310: Synthesis of \[Ru\(CO\)\(OH\)₂\(PNPtBu\)\] using Ag₂O, 5 h](#)

- [AE-325: Preparation of samples for ultrafast and raman spectroscopy from \[Ru\(CO\)\(OH\)₂\(PNPtBu\)\] \(AE-310-2\)](#)

- [AE-343: Synthesis of \[Ru\(CO\)\(OH\)₂\(PNN\)\] using Ag₂O, 5 h](#)

- [AE-352: Degassing of D₂O](#)

Linked resource

Protocol - [Inert filling of Young type cuvettes provided by AG Dietzek-Ivansic](#)

Attached files

Results-and-discussions_RuPNP_202411012.docx

sha256: 37cd85d5feaf7d0433ff0829aa5bf51c4871ce8d25c03633721db3260b5b2631

left-1-in-D2O-right-3-in-D2O.jpg

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after-filt-from-left-to-right-1-2-3-4.jpg

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AE-353.png

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AE-353.cdxml

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Unique eLabID: 20241007-e18d1c88c4d979f4224af826a214fb26c7a295de

Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=1395>